v9s Operations Guide

Day-2 KubeVirt VM Management

Practical guide for managing KubeVirt VMs after migration — console access, monitoring, snapshots, storage, network, and host administration.

For System Administrators & Platform Engineers

Console Operations

Access VMs directly from the browser — no SSH keys or VPN needed.

VNC Graphical Console

How to Access

Navigate to VMs → click VM row → Console tab → select "VNC Console"

Full graphical desktop access with mouse and keyboard. The VNC connection goes through a WebSocket proxy via virtctl. HTTPS is required for VNC.

Sendkey Toolbar

Click the keyboard icon in the VNC toolbar to reveal sendkey buttons. These send key combinations that the browser normally intercepts:

Button	Keysym	Use Case
Ctrl+Alt+Del	0xffe3 + 0xffe9 + 0xffff	Windows login, Linux SysRq, task manager
Ctrl+Alt+F1	0xffe3 + 0xffe9 + 0xffbe	Switch to text console (TTY1)
Ctrl+Alt+F2	0xffe3 + 0xffe9 + 0xffbf	Open second terminal (TTY2)
Ctrl+Alt+F7	0xffe3 + 0xffe9 + 0xffc4	Return to graphical desktop

Serial Console

When to Use

Headless VMs, boot troubleshooting, GRUB access, kernel panic debugging, VMs without graphical desktop.

SSH Console

When to Use

VMs with network connectivity and SSH server running. Fastest for command-line administration.

Monitoring & Metrics

Real-time performance data without Prometheus or Grafana setup.

Per-VM Metrics (WebSocket)

Navigate to VMs → click VM → Metrics tab. Charts update in real-time via WebSocket connection.

CPU Usage

Green area chart. Shows utilization percentage over time. 60-point rolling history. Useful for sizing and capacity planning.

Memory Usage

Blue area chart. Tracks allocation vs actual usage. Identifies memory leaks or over-provisioning.

Network I/O

Dual-line: RX (purple) and TX (amber) in KB/s. Monitor bandwidth usage and detect anomalies.

Disk I/O

Dual-line: Read (cyan) and Write (red) in KB/s. Identify storage bottlenecks and I/O spikes.

Host-Level Monitoring

Navigate to Host → Overview tab for real-time host resource usage:

- **CPU:** Usage percentage with load average (1m, 5m, 15m)
- **Memory:** Used/total with swap usage
- **Disk:** Used/total with color-coded percentage bar
- **Uptime:** Host uptime and process count

All metrics auto-refresh every 10 seconds.

Snapshots & Storage

Point-in-time backups and disk management.

Creating a Snapshot

Steps

VMs → click VM → Snapshots tab → "Create Snapshot" button → confirm

Snapshots are created via the KubeVirt VirtualMachineSnapshot API. Auto-generated name format: `vm-name-snapshot-timestamp`. The API version (v1beta1 or v1alpha1) is auto-detected.

Deleting a Snapshot

Steps

Click the trash icon next to any snapshot → confirm deletion. Loading indicator shown per item.

Mounting an ISO

Steps

VMs → click VM → Storage tab → "Mount ISO" → enter disk name and container image URL → Mount

ISOs are mounted as CDROM devices via containerDisk volumes. The mount operation modifies the VM spec with a SATA CDROM device. Duplicate disk names are automatically replaced.

Adding a Network Interface

Steps

VMs → click VM → Network tab → "Add Network" → choose name, type (pod/multus), binding (bridge/masquerade/SR-IOV) → Add

Network interfaces are added by patching the VM spec. Duplicate network names are prevented with validation. The VM may need a restart for new interfaces to take effect.

Host Administration

Five-tab host dashboard for complete system visibility.

Tab	Data Source	What You See
Overview	/proc, /sys, os-release	Hostname, OS, kernel, CPU, BIOS, vendor. CPU/memory/disk gauges with load average.
Hardware	lspci, dmidecode	RAM modules (slot, size, speed, type, vendor). PCI devices (slot, class, vendor, device).
Partitions	df, /proc/mounts	All filesystems: device, mount point, type, size, used, available. Color-coded usage bars.
Network	/sys/class/net	All interfaces: state, MAC, IPs, RX/TX bytes, speed, MTU.
Services	systemctl	All systemd services: name, status (running/exited/failed), description.

Common Host Checks

Disk Space Running Low?

Partitions tab → check usage bars. Red (>90%) means action needed. Check /var/lib/kubelet and /var/lib/containerd.

Memory Pressure?

Overview tab → memory gauge. Check swap usage. If swap is heavily used, VMs may need memory limits adjusted.

Service Failures?

Services tab → look for red "failed" badges. Common: libvirtd, k3s, v9s-web. Check with journalctl.

Network Issues?

Network tab → check interface state (up/down), IP assignments, and error counters in RX/TX.

Troubleshooting Guide

Common issues and how to resolve them with v9s.

Issue	Where to Look	Resolution
VM won't start	Events tab	Check for scheduling failures, resource limits, PVC issues
VM boots to GRUB	Serial console	Bootloader misconfigured — re-run hyper2kvm with --regen-initramfs
No network after migration	Network tab + console	Check interface binding, cloud-init network config
High CPU usage	Metrics tab	Identify the process via SSH/serial console, adjust CPU limits
Disk full on host	Host → Partitions	Clean up old images, PVCs, or snapshots
Console black screen	VNC + serial	Try serial console for headless VMs. Check VM is running.
Snapshot stuck "Pending"	Snapshots tab + Events	Check VolumeSnapshot CRDs, CSI driver status

Logs for Debugging

VM Logs

VMs → click VM → Logs tab → select "Console Log" for guest output or "Compute Log" for QEMU/libvirt messages.

v9s makes KubeVirt VM management accessible

No kubectl required. No YAML editing for daily operations.
Console, metrics, snapshots, and host info — all in one place.